

C2: Study Notes Discussion 2: Control Flow and Functions

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Overview

Extend your study notes with the concepts and tools from the control flow and structuring code chapters of our programmer's field guide, as indicated in the checklist at the end of this document.

Do the following:

- If you can, print the checklist from the end of this document and paste it into the next empty pages of your workbook. As before, use this to tick off concepts covered as you go.
- Update the checklist below as you progress.
- Move on to Reflect when you have completed all of the concepts from the checklist.

Be concise

Remember that this is about helping structure and organise your thoughts. There is no need to write a lot of text. Keep being creative, and use this to make sure you are developing and retaining the ability to engage in technical programming dialog.

Reflect:

- What is the most important thing you learned from this and why?

> I think I have deep understanding about each concept by compare it, for example is function and procedure. I also draw all concept into a huge mind map help me know about the connection between it.

- How are you keeping on top of all of the concepts and connecting them together?

> I draw the mind map to combine all term starting with Program. All repeat concept must be link, for example is expression, sequence and statement always occur in each concept like calling, or in the code block.

- Why did you select the different pages you choose to photograph and include in your portfolio?

> First page I choose the mind map that combine all the concept, I think this can represent all thing I had learned until now, so I choose it. Next I choose about the boolean algebra because is the new knowledge in this part.

Disclose:

- Explain your use of Generative AI for this task. If you didn't use it, just write "I did not use any AI for this task". If you did use it, please state which AI model you used (e.g. ChatGPT) and how.

> I did not use any AI for this task

Submit:

- Export this worksheet as a PDF.
- Take a photograph of two pages from your study notes: one that is representative of your work in general, and one that you are particularly proud of or feel was particularly influential in your learning.
- Upload your submission to OnTrack.
- Engage with feedback and discussion to get this signed off:

Checklist:

- Mind map of **structured programming** connecting the following terms (1 page):
 - Structured Programming
 - Sequence
 - Selection
 - Repetition
 - Block
- Mind map of **boolean values** connecting the following terms (1 page):
 - Data Type
 - Boolean
 - True
 - False
 - Expression
 - Variable
 - Function
- Mind map of **control flow statements** connecting the following terms (1 page):
 - Statement
 - Selection
 - Repetition
 - If Statement
 - Switch Statement
 - While Loop
 - Do While Loop
 - Boolean
 - Expression
 - Boolean Expression
- Exploration of **Boolean values** showing: (1 page)
 - Example boolean expressions
 - Different comparison operators
 - Boolean functions
 - Boolean variables

- Exploration of **Boolean operators** showing: (1+ page)
 - Example and visualisation, metaphor, or illustration for each of these:
 - And
 - Or
 - Not
 - Example boolean expressions involving multiple operator combinations with truth tables
- Exploration of the **if statement** action showing: (1 page)
 - Syntax (abstract)
 - Annotated examples of if statements
 - Details of the steps taken by the computer to perform a procedure call
 - A metaphor or illustration you connect with this action
- Mind map of **functions and procedures** connecting the following terms (1 page):
 - Program
 - Procedure
 - Function
 - Parameter
 - Call
 - Argument
 - Statement
 - Expression
- Expansion of the **procedure declaration** action showing: (1 page)
 - Syntax (abstract)
 - Annotated examples of procedure declarations
 - Description of what a procedure declaration does
 - Some example procedure ideas
- Expansion of the **function declaration** action showing: (1 page)
 - Syntax (abstract)
 - Annotated examples of function declarations
 - Description of what a function declaration does
 - Some example function ideas

- Exploration of **parameters** construct showing: (1 page)
 - Visualisation, metaphor, or illustration showing how you “see” these
 - Connect with related concepts:
 - Identifier
 - Data Type
 - Variable
 - Function / Procedure
 - Argument
 - Function Call / Procedure Call
 - Pass by value
 - Pass by reference
 - Syntax for parameter declaration / creation
 - Annotated example parameter declarations
 - Annotated example function/procedure call